INCH-POUND

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION
GENERATOR, SIGNAL, l0 KHz to l GHz

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This commercial Item description (CID), describes a laboratory standard quality bench piece of electronic test equipment which is a bench mount with rack mount option, solid-state, general purpose, 10 KHz to 1 GHz, generator signal. This CID is meant as a minimum requirement for the generator in which only those manufacturers that meet or surpass the following requirements are qualified per this CID.

2. SALIENT CHARACTERISTICS.

2.1 Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be class III in accordance with MIL-PRF-28800. This equipment shall be capable of operation within the accuracies, limits, and specifications herein.

2.2. Safety. The equipment shall be UL (Underwriters Laboratories) listed and approved or shall comply with the class III safety requirements of the latest MIL-PRF-28800.

2.2.1 Leakage current protection. Protection shall be provided from leakage current in excess of 3.5 milliamperes (ma) rms (AC and DC) from any accessible conductive parts of equipment (including control shafts with knobs broken or removed and recessed calibration or adjustment controls) to either pole of the power source for any position of the power switch(es).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC/TILCC, 420 Second Street, Suite 100, Robins AFB, GA 31098-1640.

AMSC N/A FSC 6625

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2.2.2 Radio frequency interference (RFI) capacitors. Any capacitor used in RFI filters shall be rated for at least four (4) times the operating voltage.

2.3 Design and construction. Unless otherwise specified herein, the equipment shall comply with the Class III design and construction requirements of MIL-PRF-28800. The equipment shall be of solid-state construction.

2.4. Type of construction. Construction of the generator shall be of a monolithic design.

2.4.1 Signal input and output connectors. Unless otherwise specified, all signal input and output connectors shall be type BNC female, except for RF output connector(s) which may be type N.

2.4.2 Electrical power sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 100, 120, 220 and 240 volts AC (+10%) at line frequencies of 50 Hz to 60 Hz.

2.4.3 Electrical power connectors. The equipment shall operate with a single phase input with a ground on either side.

2.4.4 Power consumption. Maximum power consumption shall not exceed 450 volt amperes V.A.

2.4.5 Fuses. Fuse(s) shall be provided, on the exterior rear panel for replacement without entering the enclosure. The fuse(s) shall automatically disconnect the equipment from the power source in the event of excessive current requirements.

2.4.6 Surge and transient voltage protection. The generator shall contain surge and transient voltage protection internal of the enclosure. The transient voltage protection circuit shall be so designed as to prevent electronic device damage within the system. The surge voltage protection circuit shall conform to IEEE Std 587-1980.

2.4.7 Dimensions and weight.

2.4.7.1 Equipment dimensions. The maximum dimensions for the generator should be approximately 9 inches height x 17 inches wide x 24 inches deep.

2.4.7.2 Equipment weight. The weight including all accessories (excluding manual) shall not exceed 90 lbs.

2.4.8 Rackmount. Rackmount capability shall be available as an option for the equipment specified herein.

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2.5 Environmental requirements.

2.5.1 Temperature requirements. Unless otherwise specified herein, the equipment shall meet its specified performance characteristics and other requirements when subjected to the class III temperature requirements specified by MIL-PRF-28800 and as listed below.
2.5.1.1 Non-operating temperature. The equipment shall meet its performance and accuracy requirements in an non-operating environment of -40°C to +71°C.

2.5.1.2 Operating temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of 0°C to 50°C.

2.5.2 Humidity requirements. The equipment shall meet performance and accuracy requirements under humidity conditions for class III equipment as specified by MIL-PRF-28800.

2.5.3 Altitude requirements. The equipment shall meet performance and accuracy requirements under the altitude conditions for class III equipment as specified by MIL-PRF-28800.

2.6. Vibration requirements. The equipment shall meet the class III vibration Requirements as specified by MIL-PRF-28800.

2.7 Mechanical shock requirements. The equipment shall meet the class III mechanical shock, requirement as specified by MIL-PRF-28800.

2.8 Electromagnetic interference (EMI). The equipment shall comply with the following EMI requirements specified in MIL-STD-461:

(1) CR01: Narrow band requirement from 10kHz to 20 kHz
(2) CSO3: Full limits.
(3) CSO1: Full limits.
(4) CSO2: Full limits.
(5) CSO6: Full limits.
(6) REO1: Relaxed 10dB at the fundamental, 3rd harmonic, and 5^{th} harmonic of the source power frequency with the measurement probe located 12 inches from device under test.
(7) REO2: Full limits to 10 times the highest used or intentionally generated frequency, or GHz, whichever is greater.
(8) RSO1: Full limits.
(9) RS03: Full limits to 10Hz

2.8.1 Mean time between failures (MTBF). The equipment mean time between failures (MTBF) shall be 2,000 hours.

2.9. Calibration and maintenance adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically

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approved. The calibration adjustments, wherever possible, shall be accessible without removal, of the instrument case or modules.

2.10 Performance characteristics. The generator shall be MATE compatible with both CIIL and ASCII controllable language through an external IEEE 488 bus configuration. The CIIL translation (TMA) shall be integral to the box where available or shall be supplied with the unit as a one unit purchase item when supplied in a separate box configuration.

2.10 Frequency. The minimum frequency range of the generator shall be 10KHz to 1GHz.

2.10.1 Resolution. The frequency resolution of the generator shall be a maximum of 1 Hz throughout the entire frequency range.

2.10.2 Accuracy. The frequency accuracy shall be based on the internal reference oscillator aging rate which shall be 4 x l0 -9 per day after 3 months of continuous operation.

2.10.3 Spectral purity.

2.10.3.1 Harmonics. The generator harmonics shall be -25 dBc or greater over the entire frequency range.

2.10.3.2 Sub-harmonics. There shall be no noticeable sub-harmonics to 1GHz.

2.10.3.3 Spurious non-harmonics. The spurious non-harmonics shall be at least -80 dBc over entire frequency spectrum.

2.10.3.4 Power line related or microphonically generated. The 60 Hz power line or 300 Hz microphonically generated effect shall be at least -80 dBc to 320 MHz and at least -70 dBc from 320 MHz to 1 GHz.

2.10.3.5 Absolute SSB phase noise. The absolute single sideband (SSB) phase noise at 200 MHz carrier at 20 KHz offset shall be at least -130 dBc per Hz.

2.11 Output. The minimum output level range into 50 ohm nominal shall be +16 dBm to -140 dBm.

2.11.1 Output resolution. The generator output resolution shall be 0.1dB.

2.11.2 Standing wave ratio. The standing wave ratio shall be less than 2.0:1 for output levels of 0 dBm or greater.

2.11.3 Output flatness. The output flatness shall be equal to or less than +0.75 dB over entire range at the +10 dbm level.

2.11.4 Output protection. The generator shall have reverse power protection of a minimum of 25W at +50 VDC from a 50 ohm source.

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2.11.5 Amplitude modulation. The amplitude modulation depth shall be 0 to 99% up to an output level of +10 dBm.

2.11.5.1 AM resolution. The AM resolution shall be 1%.

2.11.5.2 AM Accuracy. The AM accuracy at the 1 KHz rate shall be better than ± (5% of setting +1% AM).

2.11.5.3 Distortion. The generator AM distortion at the 1 KHz rate shall be less than 2% up to 30% AM and less than 6% up to 80% AM.

2.11.5.4 Incidental phase modulation. The incidental phase modulation at the 1 KHz rate and 30% AM shall be less than 0.2 radians peak.

2.11.6 Frequency modulation and coupling. The modulation bandwidth at 3 dB down shall be a minimum of 20 Hz to 100 kHz AC coupled. DC to 100 kHz DC coupled internal or external into 600 ohms.

2.11.6.1 Incidental AM with FM. The incidental AM with FM shall be less than 0.3% for a deviation of 20 KHz at a 1 KHz rate.

2.11.6.2 FM distortion. The generator shall have a distortion of less than 4% for maximum do coupled deviation and less than 2% for 50% maximum deviation.

2.11.6.3 Accuracy. The FM indicator accuracy shall not exceed ± (5% of setting +10 Hz) maximum.

2.11.6.4 Resolution. The FM resolution shall be less than 1.0% of setting or 0.0005% of maximum deviation whichever is larger.

2.11.6.5 Deviation. The maximum deviation at 1 GHz shall be DC to 100 KHz.

2.11.7 Phase modulation. The maximum phase deviation at 1 GHz shall not be greater than 200 radians.

2.11.7.1 Accuracy. The deviation accuracy shall not be greater than 10% at the 1 KHz rate.

2.11.7.2 Bandwidth. The minimum 3 dB phase modulation bandwidth shall be 20 Hz to 10 KHz.

2.11.7.3 Resolution. The synthesized signal generator phase modulation resolution shall be within 0.8% of setting or 0.005% of maximum deviation, whichever is greater.

2.11.8 Pulse modulation. The on/off ratio for the pulse modulation shall be 40 dB or greater.

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2.11.8.1 Rise and fall time. The rise and fall time between 10% and 90% of the pulse shall be 0.4 microseconds or less.

2.11.8.2 Repetition rate. The minimum repetition rate for a carrier frequency of 10 MHz to 1 GHz shall be 20 Hz to 100 KHz AC and DC to 100 KHz DC.

2.11.8.3 Pulse width. The minimum pulse width shall be at least 2 microseconds.

2.12 Input level. The nominal peak input threshold shall be 1.5V.

2.13 Internal modulation. The generator shall have an internal modulation source of a minimum of 400 Hz and 1 KHz with a minimum output level of 2V rms +0.5 dB into 600 ohms.

2.14 Digital sweep. The generator shall have a variable 10 KHz to 1 GHz front panel selectable sweep with selectable sweep speed.

2.15 Manuals. The unit shall come with two(2) complete sets of operation, instruction and programming manuals.

2.16 Calibration interval. After calibration, the equipment shall meet each performance requirement within the tolerances specified for a period of at least 9 months, based on an operating time of 1500 hours.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5. PACKAGING.

5.1 Preservation, packaging, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

Federal and Military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

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IEEE Std 587-1980. Application for copies of IEEE publications should be addressed to the Institute of Electrical and Electronics Engineers, Inc., 345 East 47th Street, New York, NY 10017.

6.2 Ordering data. Procurement documents must specify the title, number, and date of this CID, the issue of the DODISS to be used in the solicitation, and preservation and packaging requirements.

Custodian:
Navy – EC
DLA -CC

Review activity:
Army - CR

Preparing activity:
Air Force - 84

Agent:
Air Force - 99

Project Number 6625—0908

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